

AI & Spring AI – Complete FAQ Guide

Section 1: Artificial Intelligence Basics

Q: What is Artificial Intelligence (AI)?

A: Artificial Intelligence is the simulation of human intelligence in machines that are programmed to think, learn, and make decisions.

Q: What is Machine Learning?

A: Machine Learning is a subset of AI that enables systems to learn from data and improve performance without being explicitly programmed.

Q: What is Deep Learning?

A: Deep Learning is a subset of Machine Learning that uses neural networks with multiple layers to model complex patterns in data.

Q: What is Natural Language Processing (NLP)?

A: Natural Language Processing is a field of AI focused on enabling machines to understand, interpret, and generate human language.

Q: What is a Large Language Model (LLM)?

A: A Large Language Model is a deep learning model trained on vast amounts of text data to understand and generate human-like language.

Q: What is an embedding?

A: An embedding is a numerical representation of text, image, or other data in vector form so that similar items are closer together in vector space.

Q: What is a vector database?

A: A vector database is a specialized database designed to store and query vector embeddings efficiently using similarity search algorithms.

Q: What is Retrieval-Augmented Generation (RAG)?

A: Retrieval-Augmented Generation is an architecture that combines information retrieval with text generation to provide more accurate and context-aware responses.

Q: What is prompt engineering?

A: Prompt engineering is the practice of designing effective prompts to guide AI models toward generating accurate and relevant outputs.

Q: What is AI hallucination?

A: Hallucination in AI refers to when a language model generates information that appears correct but is factually incorrect or fabricated.

Q: What is fine-tuning?

A: Fine-tuning is the process of training a pre-trained model on a smaller, domain-specific dataset to improve its performance on specialized tasks.

Q: What is transfer learning?

A: Transfer learning is a technique where a model trained on one task is reused as the starting point for a related task.

Q: What is a transformer model?

A: A transformer model is a neural network architecture that uses self-attention mechanisms to process sequential data efficiently.

Section 2: Spring AI Basics

Q: What is Spring AI?

A: Spring AI is a framework that integrates Artificial Intelligence capabilities into Spring-based applications.

Q: How does Spring AI integrate with LLMs?

A: Spring AI integrates with Large Language Models using abstractions like ChatClient and embedding models.

Q: What is ChatClient in Spring AI?

A: ChatClient is a Spring AI component used to interact with language models for generating conversational responses.

Q: What is VectorStore in Spring AI?

A: VectorStore is used in Spring AI to store embeddings and perform similarity search for retrieval-based applications.

Q: How does Spring AI support RAG?

A: Spring AI supports Retrieval-Augmented Generation by combining VectorStore retrieval with language model generation.

Q: What is PromptTemplate in Spring AI?

A: PromptTemplate is used to dynamically generate structured prompts by injecting context variables into predefined templates.

Q: How does metadata filtering work in Spring AI?

A: Metadata filtering allows filtering retrieved documents based on metadata fields during similarity search.

Q: What are common use cases of Spring AI?

A: Common use cases include chatbots, knowledge assistants, document search systems, and intelligent automation.